

Vishal Tak

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EDUCATION

The University of Texas at Austin — Austin, TX

May 2022

Bachelor of Science in Computer Science – GPA: 3.72

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, Apex

Frameworks & Libraries: Spring Boot, Spring Batch, JPA/Hibernate, Angular, React, Salesforce Lightning Web Components (LWC), REST APIs, Cypress, Spock, JUnit

AI / ML: LLMs, Retrieval-Augmented Generation (RAG), Vector Databases, Spec-Driven Development (SDD)

Cloud & Infrastructure: AWS (Fargate/ECS, Batch, Lambda, S3, IAM, EventBridge), Docker, CI/CD, OpenTelemetry (OTEL), New Relic, Splunk

Databases & Storage: PostgreSQL, Snowflake, Amazon Neptune (Gremlin)

Certifications & Key Highlights: U.S. Patent Granted (SQL Query Combiner) + 1 Filed • Promoted 2x in < 3.5 Years (Associate to Senior SWE) • AWS Certified Solutions Architect (SAA-C03)

PROFESSIONAL EXPERIENCE

Capital One — Plano, TX

August 2022 – Present

Senior Software Engineer

Jan 2026 – Present

- Replaced fragmented Slack and email approvals by architecting an event-driven approval microservice (Spring Boot, PostgreSQL, AWS Fargate) with an idempotent state machine and dynamic task routing, now centralizing 100+ approvals/day across 5+ use cases and 300+ sales agents.
- Built an AI-driven observability workflow that generates New Relic dashboards and alerts on demand, cutting metric setup from days to minutes, drastically improving developer efficiency and earning Runner-Up at a company hackathon.
- Identified and fixed a state-management bug in the core exception-limit business logic that allowed a rolling-window lockout to reset improperly, letting users bypass usage limits, and restored correct enforcement in production with zero regressions.

Software Engineer

Jan 2024 – Jan 2026

- Led the cross-team design and delivery of a new dealer-compensation platform that pays dealers a higher per-loan rate for hitting monthly targets, increasing dealer payouts and loan volume across the dealer network.
- Built and owned the production-grade AWS Spring Batch component (IAM, job queues, EventBridge, retry logic, Splunk alerting) that automated the monthly compensation calculations and ensured its on-time launch.
- Resolved a customer-facing production incident by rerouting a stale batch-fed endpoint to an hourly-updated source and driving the postmortem that retired the failing upstream job, earning an internal engineering excellence award.
- Increased dealer awareness of benefits by 8x by building a dealer-benefits MFE backend endpoint with a dynamic, tier-keyed messaging engine.
- Reduced developer debugging time 10x and boosted legacy-to-next-gen output parity by 20% by designing a Python dependency-tree engine that traces complex variable lineage across distributed decision graphs.
- Orchestrated distributed, fault-tolerant execution of ~1,000-node Directed Acyclic Graphs (DAGs) spanning six microservices on the core loan-decisioning platform, processing 80K+ loan applications per day with zero execution failures.

Associate Software Engineer

Aug 2022 – Jan 2024

- Slashed decisioning-engine execution latency by 50% by refactoring single-threaded graph processing into a multi-threaded worker pool with Java concurrency primitives (ExecutorService, CompletableFuture).
- Built the graph-authoring and simulation experience on a stack new to me, developing an Angular/TypeScript UI over Spring Boot services and an Amazon Neptune (Gremlin) graph database, with topological sorting to structure and visualize DAG execution.

PROJECTS & LEADERSHIP

- **MFE Factory (AI-Assisted):** Engineered an AI-driven Figma-to-code SDD (Spec Driven Development) workflow that improved micro-frontend (MFE) delivery efficiency by ~90%, adopted across 10+ engineering teams to standardize UI architecture and component reuse.
- **GenAI Account-Planning Assistant:** Built a Retrieval-Augmented Generation (RAG) assistant (a Python orchestrator over a vector database and an LLM) that automates the manual monthly account-planning workflow for sales agents, turning fragmented CRM and reporting data into actionable insights.
- **Tech Pro-Bono (Mobile & Nonprofit):** Built a React Native mobile-app MVP for an early-stage startup and automated SMS deadline reminders for a college-access nonprofit using Salesforce automation, reducing manual outreach effort.
- **CODERS Mentorship:** Mentor middle-school students across multi-week programs, teaching HTML/CSS and app development (MIT App Inventor) and guiding them to build their own projects.